UED IA2

Color Choices in UI: Impact on Accessibility for Color Blind Users

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1. Problem Definition

Color is a fundamental element in modern **User Interface (UI) design**, used to create hierarchy, communicate status, and provide visual feedback. However, when designers rely solely on color to convey meaning, such as red for errors or green for success, it creates accessibility barriers for users with **Color Vision Deficiency (CVD)**.

Color blindness, including **Deuteranopia** (green deficiency), **Protanopia** (red deficiency), and **Tritanopia** (blue-yellow deficiency), affects how users perceive and differentiate colors. For such users, color-coded elements like buttons, alerts, or status indicators may appear indistinguishable.

For example, a red "Cancel" button and a green "Confirm" button can look similar to color-blind users, leading to confusion or unintended actions. Similarly, when order statuses or offers are indicated using only color, users may misinterpret the meaning, resulting in reduced usability and poor user experience.

The identified problem is the lack of accessible and inclusive color design in many interfaces, which affects both perceivability and understandability for color-blind users. The project therefore focuses on improving UI color accessibility through contrast correction, meaningful color usage, and alternative visual cues in line with **WCAG 2.1** guidelines.

2. Accessibility & Inclusivity Principles Implemented

The project was guided by Accessibility and Inclusivity principles derived from WCAG 2.1 (Web Content Accessibility Guidelines) to ensure equal usability for all users, regardless of their visual perception.

Accessibility Principles Applied:

- **Perceivable:** Information and UI elements must be visible and distinguishable, even without color cues.
- **Operable:** All interactive components should remain functional and visually identifiable.
- **Understandable:** Colors, text, and icons must consistently convey meaning.
- **Robust:** The interface should remain compatible with various assistive tools.

Inclusivity Principles Applied:

- Designing for users with Deuteranopia (green deficiency), Protanopia (red deficiency), and Tritanopia (blue-yellow deficiency).
- Ensuring equal user experience by combining text, color, and iconography.
- Creating an environment where no visual information is dependent on color alone.

3. Discussion – Addressing End-User Challenges

Type	Description	Effect on UI / Perception
Deuteranopia	Green color blindness	Red and green appear similar; green tones may look brownish.
Protanopia	Red color blindness	Red appears dull or gray; hard to distinguish from dark greens.
Tritanopia	Blue–yellow blindness	Blue and yellow hues appear faded or interchangeable.
Deuteranomaly	Weak green perception	Green appears more red; mild confusion between reds and greens.
Protanomaly	Weak red perception	Reds look darker; may confuse red with brown or black.
Monochromacy (Total Color Blindness)	No color perception at all	The entire interface seen in grayscale, relies fully on brightness contrast.

Users with color vision deficiencies face challenges such as:

- Difficulty distinguishing between action buttons (e.g., *Add to Cart* vs *Cancel*).
- Inability to identify color-based status indicators (e.g., restaurant open/closed, order status).
- Poor readability due to low contrast between text and background.

How the Implemented Principles Address These Issues:

- Introduced **text + icon + color** for all visual indicators to eliminate color dependency.
- Used **high-contrast color palette** meeting WCAG 2.1 AA contrast ratio (≥4.5:1).
- Ensured **consistent color coding** throughout the UI (e.g., red for errors, blue for primary actions, green for success).
- Added **focus outlines and readable typography** for improved visual clarity.

This approach ensures that both color-blind and non-color-blind users can perceive, operate, and understand all interface elements equally.

Color-Blind Type	Challenge	Solution Implemented
Deuteranopia (Green Deficiency)	Cannot distinguish red/green	Added ✅/❌ labels + icons
Protanopia (Red Deficiency)	Red appears dull or gray	Textual cues replace color dependence
Tritanopia (Blue–Yellow Deficiency)	Confusion in blue/yellow tones	Used distinct luminance + text contrast
Low Vision	Low color contrast readability	Adopted WCAG-approved color contrast

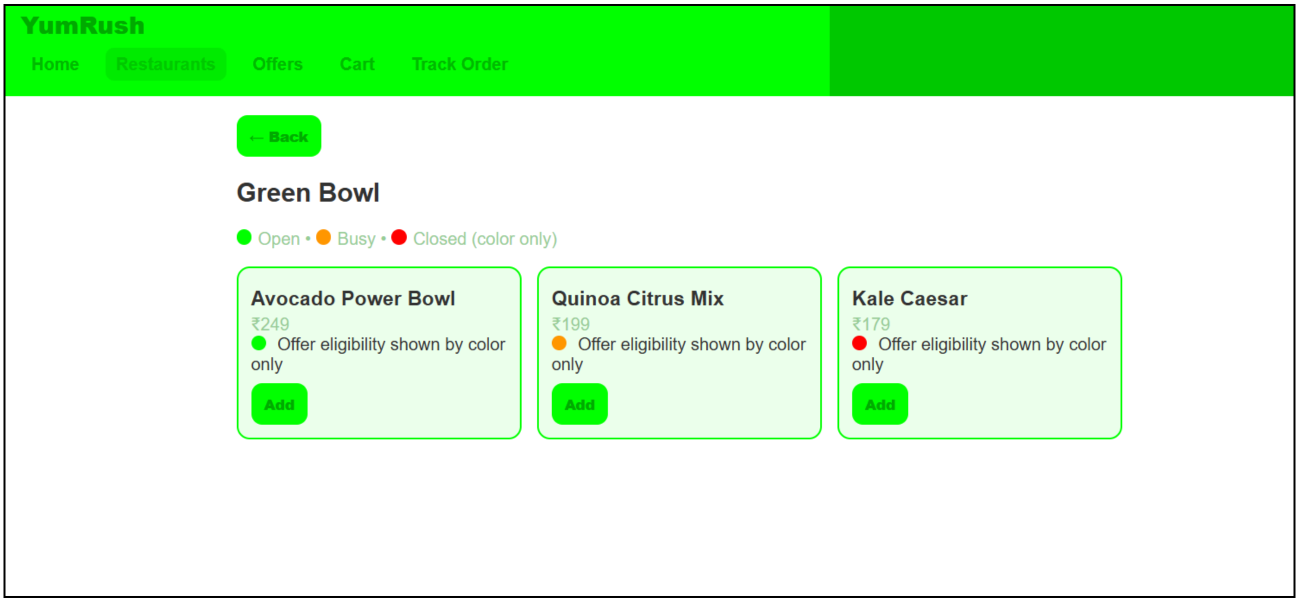
4. Implementation Techniques and Methodology

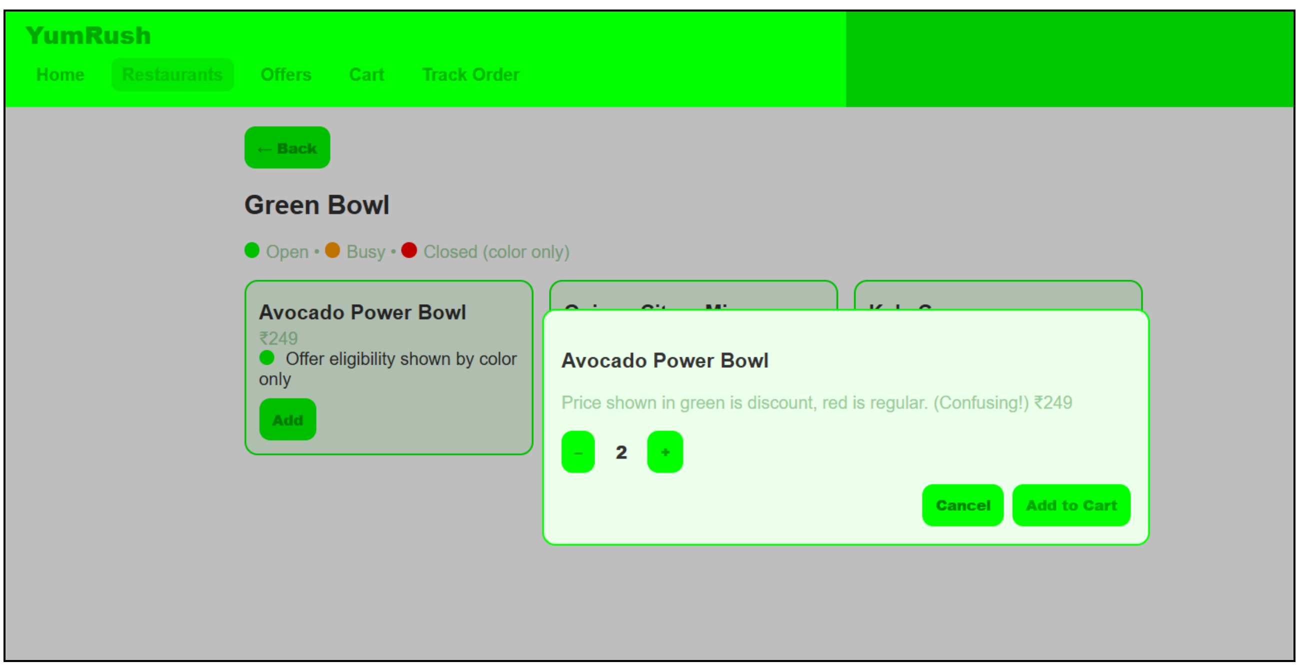
Application Chosen:
A Food Ordering Web App (YumRush) was selected to model accessibility improvements in a real-world scenario where colors indicate actions, offers, and statuses.

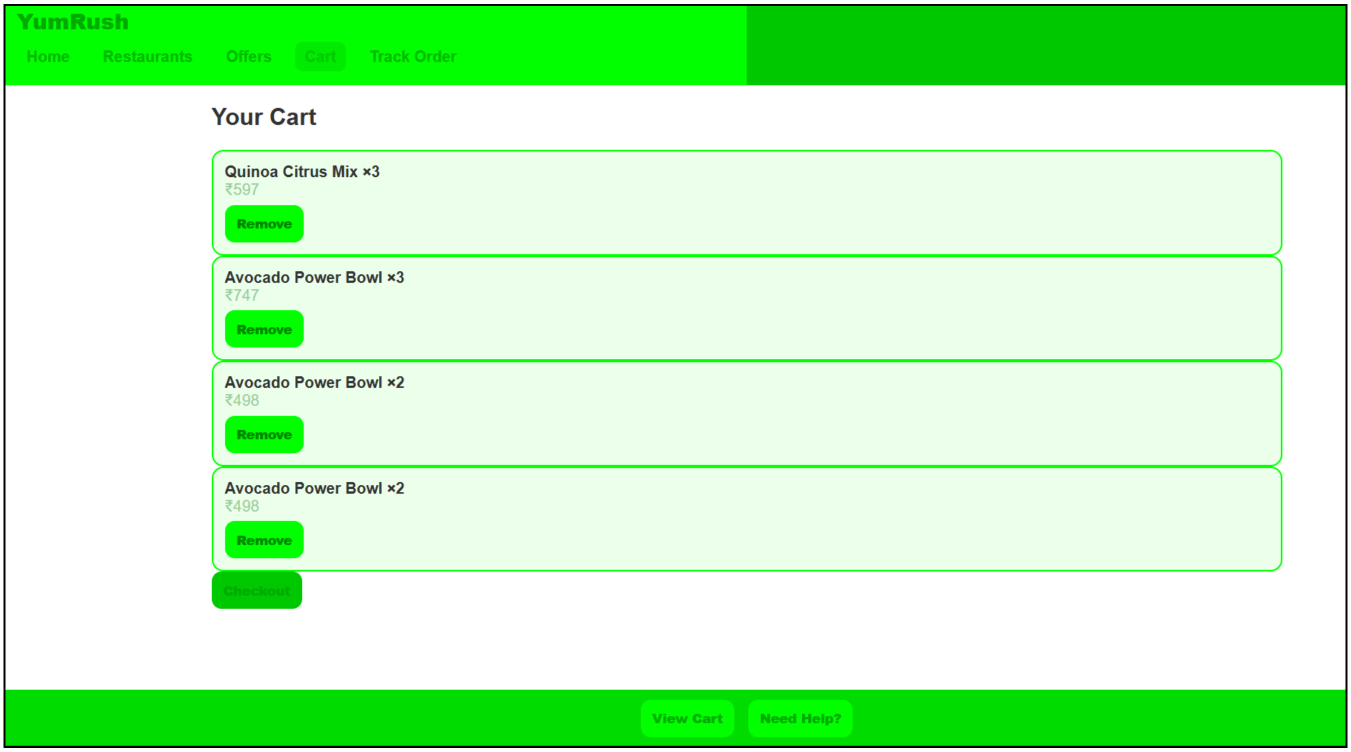
Methodology:

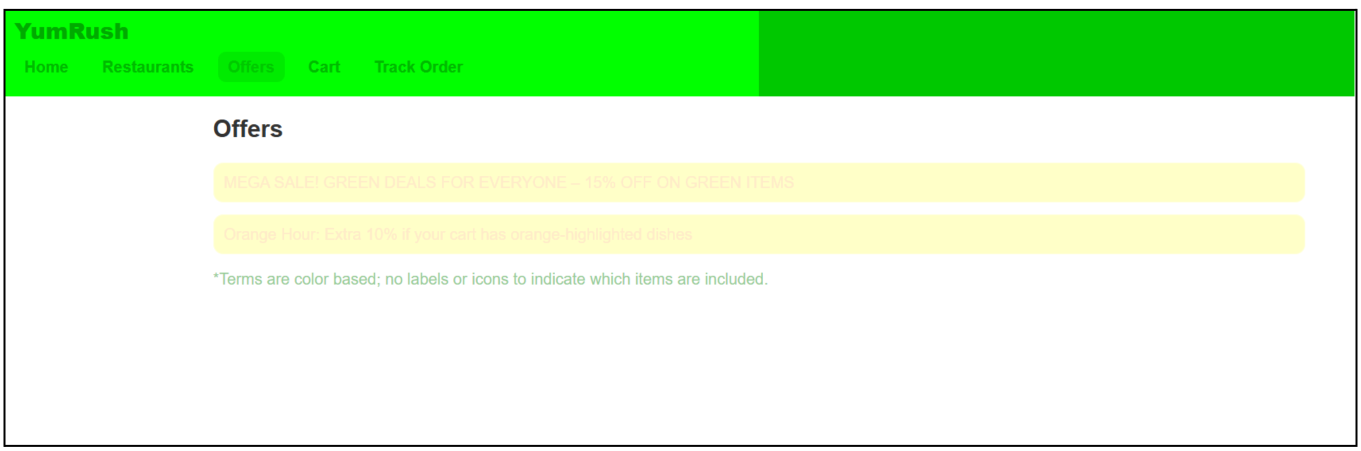
- **Phase 1 – Analysis:** Identified non-accessible design elements such as low-contrast text, inconsistent color use, and color-only indicators.
- **Phase 2 – Design:** Created two prototypes —
  - **Before (Non-accessible):** Poor color contrast, confusing button colors, and reliance on color-only cues.
  - **After (Accessible):** Implemented WCAG 2.1-compliant colors, textual indicators, icons, and improved contrast.
- **Phase 3 – Development:** Built the prototypes using HTML, CSS, and JavaScript to simulate interactive behavior such as menu navigation, cart management, and order tracking.
- **Phase 4 – Comparison:** Conducted visual analysis between the two prototypes to demonstrate the improvement in accessibility and inclusivity.

1) Before Prototype Snapshots:

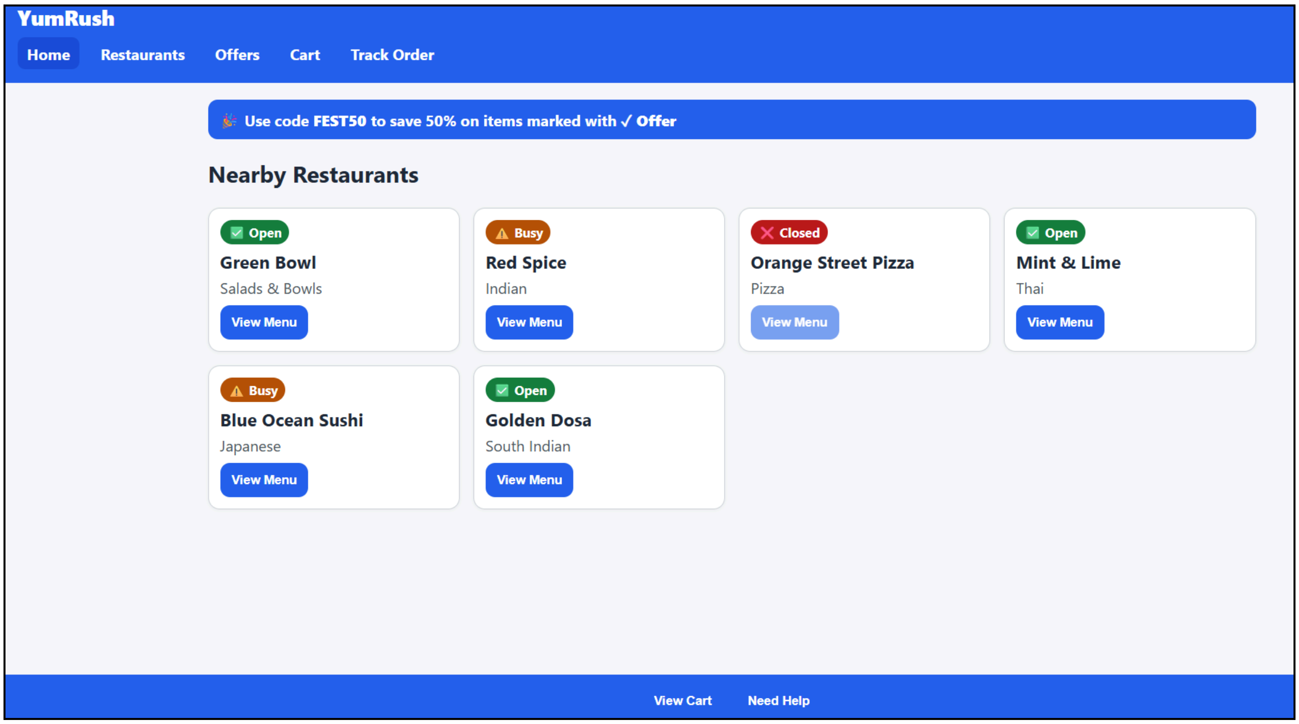


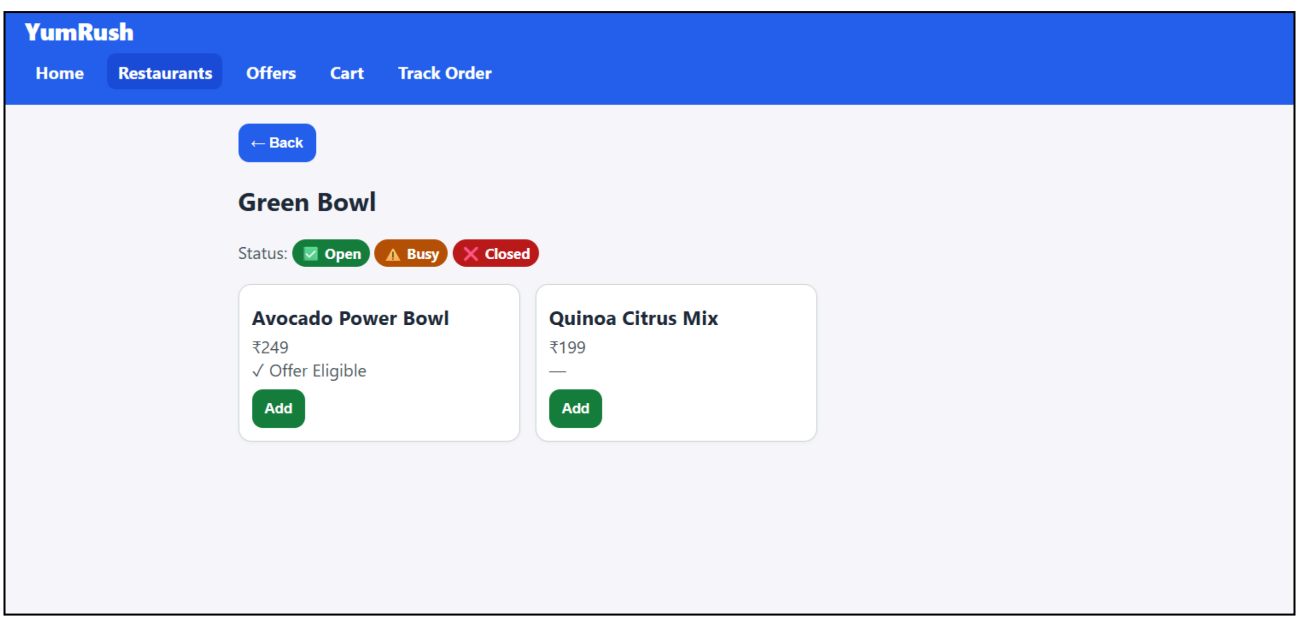


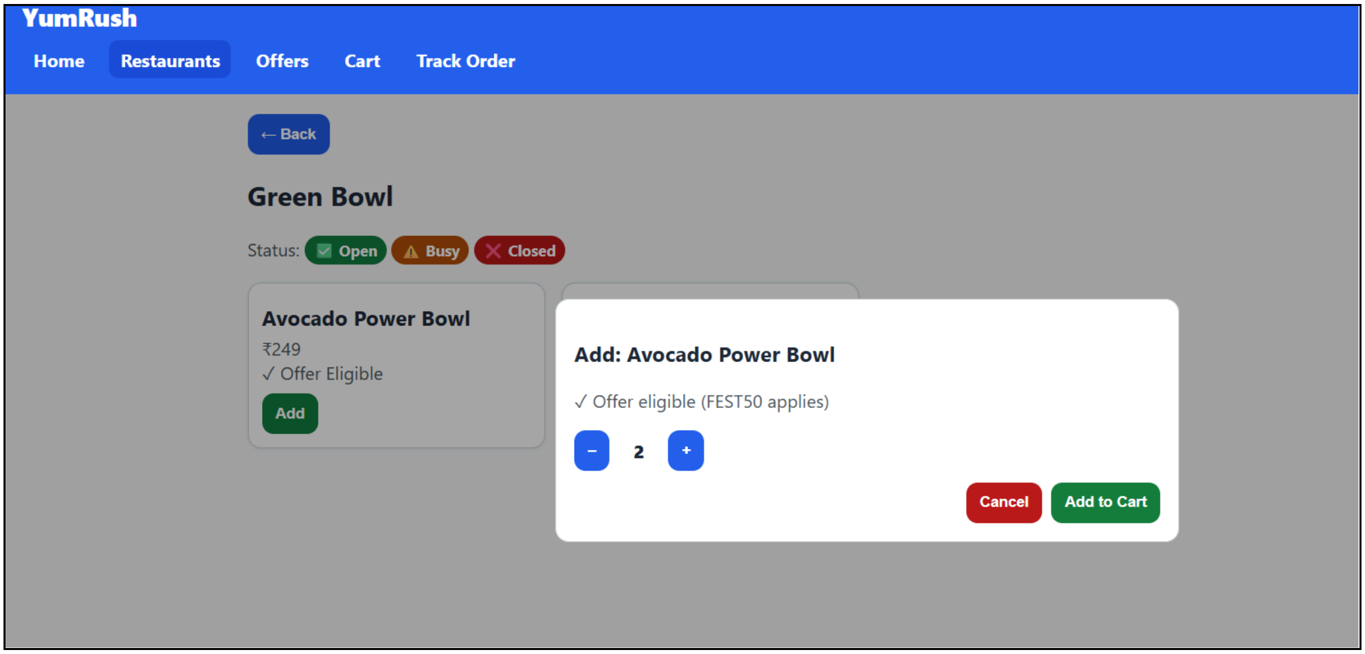

YumRush
Home
Restaurants
Offers
Cart
Track Order
Your Cart
Quinoa Citrus Mix ×3
₹597
Remove
Avocado Power Bowl ×3
₹747
Remove
Avocado Power Bowl ×2
₹498
Remove
Avocado Power Bowl ×2
₹498
Remove
Checkout
View Cart
Need Help?

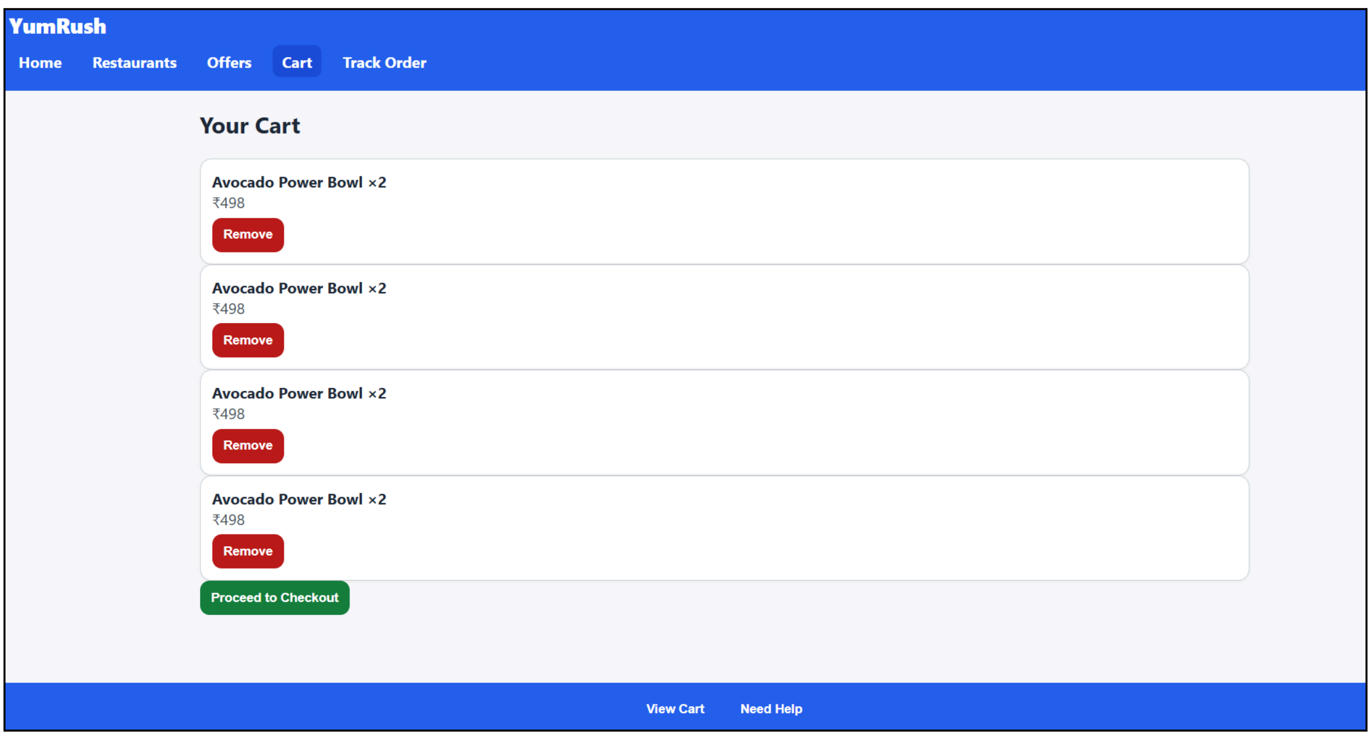

YumRush
Home
Restaurants
Offers
Cart
Track Order
Offers
MEGA SALE! GREEN DEALS FOR EVERYONE – 15% OFF ON GREEN ITEMS
Orange Hour: Extra 10% if your cart has orange-highlighted dishes
*Terms are color based; no labels or icons to indicate which items are included.

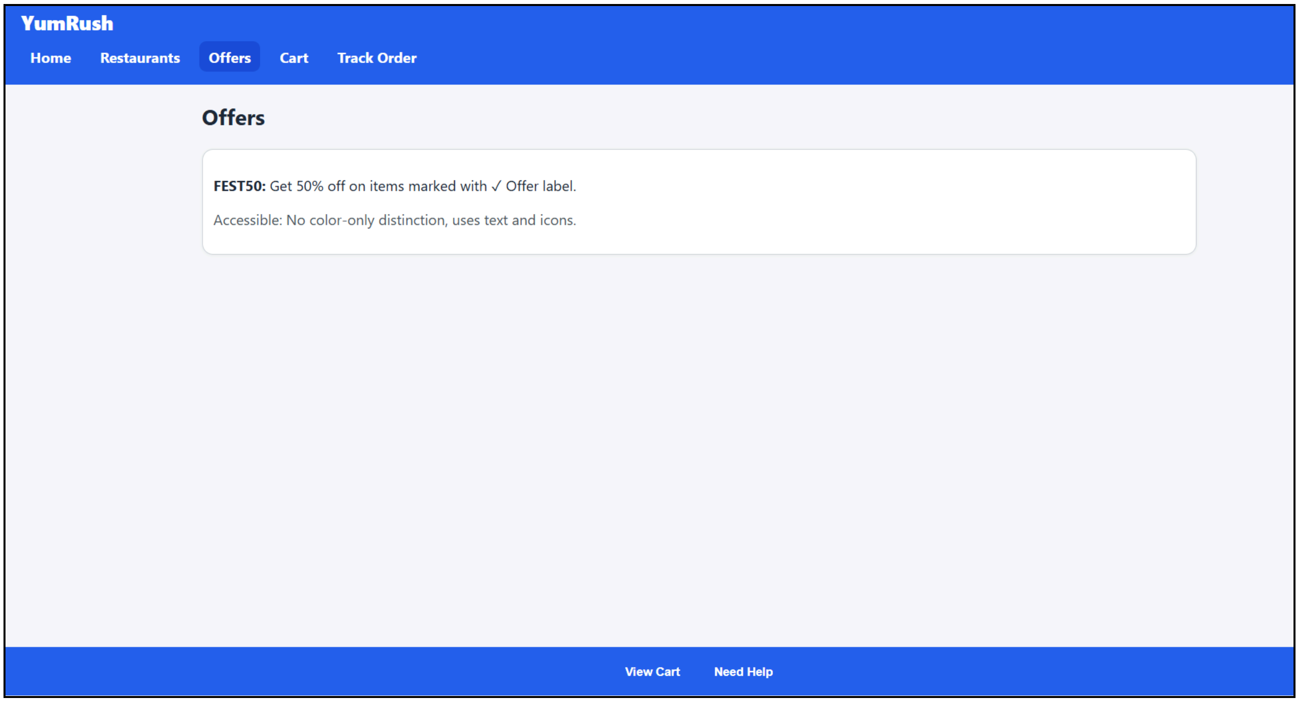
2. After Prototype






YumRush
Home
Restaurants
Offers
Cart
Track Order
← Back
Green Bowl
Status: Open Busy Closed
Avocado Power Bowl
₹249
✓ Offer Eligible
Add
Add: Avocado Power Bowl
✓ Offer eligible (FEST50 applies)
– 2 +
Cancel
Add to Cart


YumRush
Home
Restaurants
Offers
Cart
Track Order
Your Cart
Avocado Power Bowl ×2
₹498
Remove
Avocado Power Bowl ×2
₹498
Remove
Avocado Power Bowl ×2
₹498
Remove
Avocado Power Bowl ×2
₹498
Remove
Proceed to Checkout
View Cart
Need Help


YumRush
Home
Restaurants
Offers
Cart
Track Order
Offers
FEST50: Get 50% off on items marked with ✓ Offer label.
Accessible: No color-only distinction, uses text and icons.
View Cart
Need Help

5. Prototype Results

Parameter	Before Prototype	After Prototype
Color Usage	Relied on color only	Combined text + icon + color
Contrast Ratio	< 3:1 (low contrast)	≥ 4.5:1 (WCAG 2.1 compliant)
Button Consistency	Confusing (same color for Cancel/Remove)	Distinct and meaningful colors
Readability	Low, difficult for color-blind users	Clear and perceivable by all
Accessibility Standard	Not compliant	WCAG 2.1 AA compliant
User Inclusivity	Excluded CVD users	Inclusive and universally accessible

The **Before** version highlighted usability gaps such as green "Cancel" buttons and similar shades for multiple actions.
The **After** version resolved these through **high-contrast colors**, **icon indicators**, and **text labels**, creating a more intuitive and accessible interface.

6. Conclusion

The project clearly demonstrates how implementing **color accessibility and inclusive design principles** can make user interfaces more usable and equitable for all users. By addressing the challenges faced by individuals with **color vision deficiencies (CVD)**, the redesigned prototype promotes better understanding, clarity, and interaction across the interface.

Following **WCAG 2.1 guidelines**, the improved design integrates **color, text, and icon combinations** to communicate meaning effectively, ensuring that users can interpret information without relying on color alone. The results show that simple yet thoughtful adjustments—like improving contrast and providing alternative visual cues—can significantly enhance accessibility.

This study highlights that accessibility is not an optional feature but a **core component of effective UX design**. When inclusivity is prioritized, interfaces become more intuitive, visually balanced, and user-friendly for everyone. Accessible design ultimately improves **usability, satisfaction, and trust**, reinforcing that good design is inclusive by nature.

7. References

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[2] WebAIM (2024). *Contrast Checker – Evaluate Color Contrast for WCAG Compliance.*
→ https://webaim.org/resources/contrastchecker/

[3] Coblis (2024). *Color Blindness Simulator – Visualize Color Deficiency.*
→ https://www.color-blindness.com/coblis-color-blindness-simulator/

[4] Nielsen Norman Group (NNG) (2023). *Using Color to Enhance Your Design.*
→ https://www.nngroup.com/articles/color-enhance-design/

[5] Smashing Magazine (2024). *A Practical Guide to Designing for Colorblind People.*
→ https://www.smashingmagazine.com/2024/02/designing-for-colorblindness/